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Class:- BSCS 3F

Roll No:- SU92-BSCSM-F25-324

Instructor:- Sir Hmaza

Home Tasks:- Lab 06

Home Task 1:

Small and Capital Alphabet using AND

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX


    MOV CX, 26
    MOV DL, 'a'

SMALL:
    MOV AH, 02H
```

```
    INT 21H
    INC DL
    LOOP SMALL

    MOV DL, 0DH
    MOV AH, 02H
    INT 21H

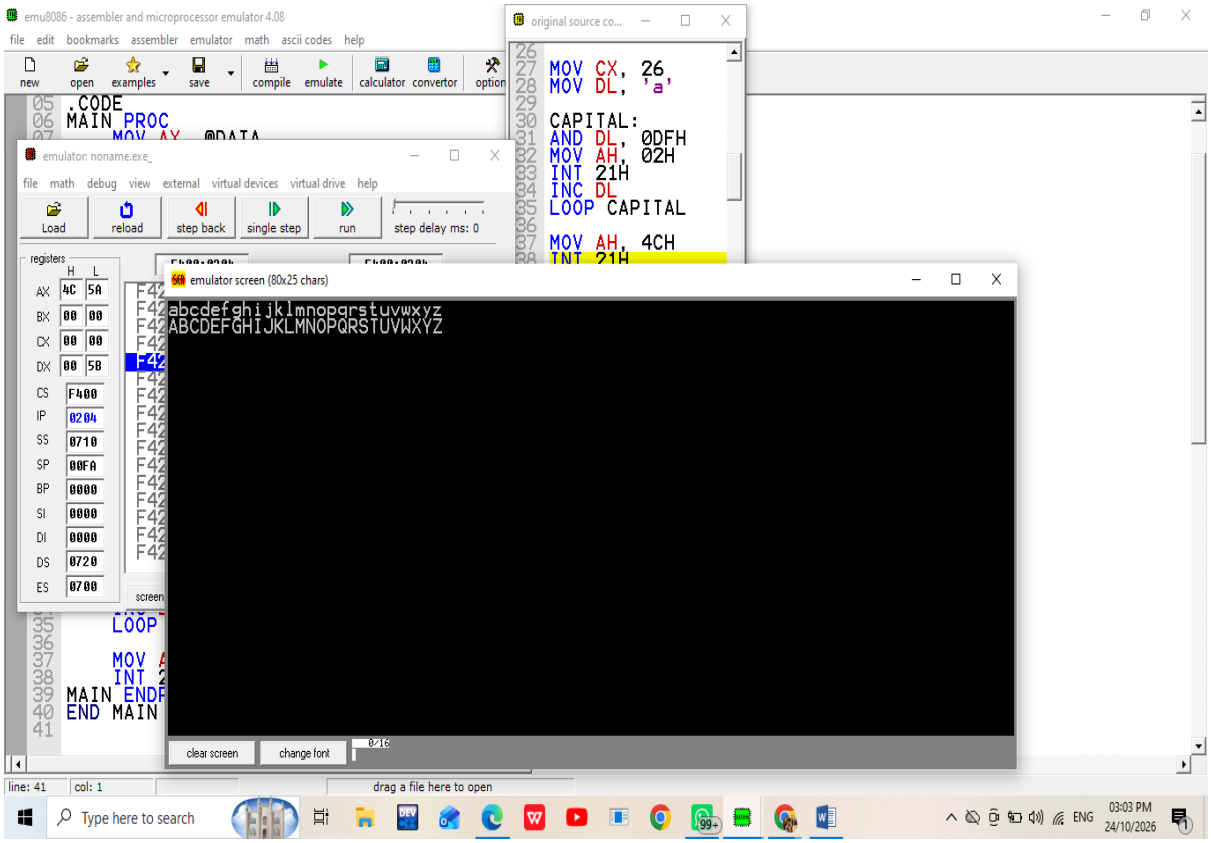
    MOV DL, 0AH
    MOV AH, 02H
    INT 21H

    MOV CX, 26
    MOV DL, 'a'

CAPITAL:
    AND DL, 0DFH
    MOV AH, 02H
    INT 21H
    INC DL
    LOOP CAPITAL

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output:-



Home Task 2:

Convert Typed Letters to Capitals

```
.MODEL SMALL
.STACK 100H
.DATA
    LETTERS DB 5 DUP(?)
    MSG DB 'Enter 5 small letters: $'
    MSG2 DB 'Converted: $'

.CODE
```

```
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    LEA DX, MSG
    MOV AH, 09H
    INT 21H

    MOV CX, 5
    MOV SI, 0

INPUT:
    MOV AH, 01H
    INT 21H
    MOV LETTERS[SI], AL
    INC SI
    LOOP INPUT

    MOV DL, 0DH
    MOV AH, 02H
    INT 21H

    MOV DL, 0AH
    MOV AH, 02H
    INT 21H

    LEA DX, MSG2
    MOV AH, 09H
    INT 21H

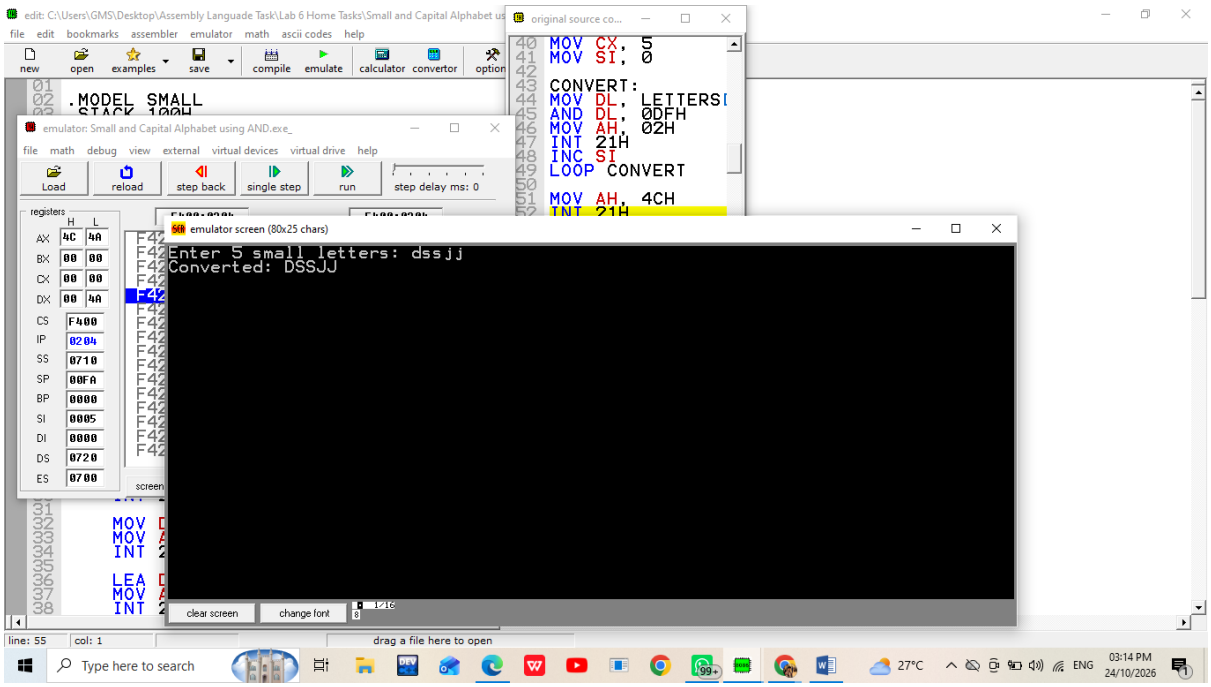
    MOV CX, 5
```

```
    MOV SI, 0

CONVERT:
    MOV DL, LETTERS[SI]
    AND DL, 0DFH
    MOV AH, 02H
    INT 21H
    INC SI
    LOOP CONVERT

    MOV AH, 4CH
    INT 21H
MAIN ENDP
END MAIN
```

Output:-



Home Task 3:

Even Digits Using AND

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    MOV CX, 10
    MOV BL, 9

CHECK:
    MOV DL, BL
    AND DL, 01H
    JNZ SKIP

    MOV DL, BL
    ADD DL, 30H
    MOV AH, 02H
    INT 21H

SKIP:
    DEC BL
    LOOP CHECK

    MOV AH, 4CH
```

INT 21H

MAIN ENDP

END MAIN

Output:-

